
Temporal Graph Learning for Point-in-Time Crypto Correlation Forecasting

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Abstract

Crypto assets trade continuously and often become highly correlated during market stress, making covariance estimation central to risk allocation. I study an edge-regression problem: given a point-in-time liquidity universe of the top 50 Yahoo crypto symbols, predict pairwise future realized correlations from lagged returns, volatility, and graph context. I compare persistence, EWMA, residual histogram gradient boosting (HGB), tabular residual MLP and all-node Transformer baselines, Temporal Graph Attention Networks (GAT), a softmax-correlation GAT variant, and an edge-temporal GRU. The learned residual and neural models beat persistence and EWMA on RMSE; the best attention models also dominate the finance baselines in downstream fast TSMOM Sharpe. Residual HGB is the strongest pointwise forecaster, while Temporal GAT and all-node attention often produce better downstream allocation. Matrix-level diagnostics suggest why: models with graph-wide structure better capture the common crypto market movement across coins, whereas models without explicit graph interaction tend to underestimate this dominant shared mode when it matters most. The results suggest that average correlation RMSE and portfolio usefulness are related but not identical objectives.

1 Introduction

Problem. For each active asset pair (i, j) at date t , predict future realized correlation $\rho_{ij,t}^{\text{fwd}}$ over either 60 or 120 days. Models output a residual over a current rolling baseline:

$$\hat{\rho}_{ij,t} = \rho_{ij,t}^{\text{base}} + f_{\theta}(x_{ij,t}, G_t), \quad y_{ij,t} = \rho_{ij,t}^{\text{fwd}} - \rho_{ij,t}^{\text{base}}.$$

The learning target is therefore an edge-level continuous label on a dynamic graph whose nodes are coins and whose edges are pairwise dependence estimates.

Motivation. Correlation forecasts matter because systematic crypto strategies can be badly mis-sized when assets suddenly move together. A model that slightly improves average pairwise correlation error may still be unhelpful if its errors occur on low-weight pairs; conversely, a graph model may help if it captures market-wide relation structure that matters for covariance-scaled allocation.

Summary. Learned residual and neural models improve materially over persistence and EWMA baselines in forecast RMSE, and attention-based neural models are the strongest downstream fast TSMOM allocators. Residual HGB achieves the best RMSE for both horizons (0.198 for 60d, 0.162 for 120d). However, downstream fast time-series momentum Sharpe is highest for hard-threshold GAT in both experiments (1.27 for 60d and 1.29 for 120d). This discrepancy is the core empirical finding: the best pointwise correlation model is not always the best portfolio model.

2 Dataset and EDA

I use Yahoo Finance daily close and volume data, constructing a point-in-time (PIT) monthly universe: the top 50 symbols by trailing 90-day median Yahoo dollar volume, excluding stablecoins, wrapped

tokens, duplicates, and obvious data-quality failures. The cleaned return panel spans 2020-09-28 to 2026-04-26. The PIT universe contains 185 unique symbols over time, with exactly 50 active symbols per day.

Returns are daily log returns. Forecast features use overlapping 3-day log returns $R_i[t] = r_i[t] + r_i[t-1] + r_i[t-2]$. For the 60d pair panel there are 2.21M pair-date rows and 7,913 unique active pairs. The thresholded graph with $|\rho^{\text{base}}| \geq 0.5$ has mean 846 edges, density 0.725, average degree 34.7, and average clustering coefficient 0.865. Future correlations are mostly positive: 95.1% of 60d labels are above 0.2, 4.7% are neutral, and 0.2% are below -0.2.

Exploratory Data Analysis: Yahoo PIT Monthly Liquidity Top-50

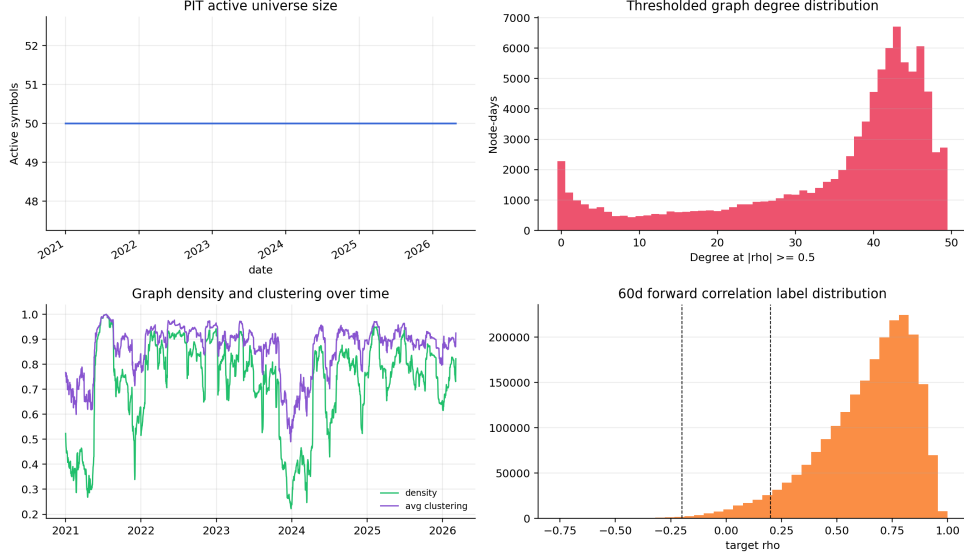


Figure 1: EDA from the PIT liquidity universe. The crypto graph is dense and strongly clustered; future correlation labels are heavily positive, so RMSE and rank metrics are more informative than sign classification alone.

Splits. Each fold uses 720 pre-test days. Tuned models use 600 chronological training days and 120 validation days; the next 60 dates are held out for forecasts. The 60d experiment has 19 folds through 2026-02-03; the 120d experiment has 18 folds through 2025-12-05 because longer labels require more future data. Hyperparameters are selected within each fold only.

3 Related Work

Classical dynamic conditional correlation models estimate time-varying covariance with strong financial interpretability [Engle, 2002]. They succeed as stable risk models and motivate the persistence/EWMA baselines here. However, they are parametric, hard to scale to a changing 50-asset PIT universe, and do not naturally use coin-level or pair-level covariates. My project keeps their core idea of rolling dependence estimation but learns residual corrections from richer features.

Covariance-shrinkage work also argues that naive sample covariance estimates are too noisy for portfolio optimization [Ledoit and Wolf, 2004]. That paper is relevant because it frames risk forecasting as an input-estimation problem: better covariance estimates can improve downstream portfolio decisions even without changing expected-return signals. My project follows the same motivation but replaces linear shrinkage with learned correlation residuals and evaluates them through ERC and TSMOM allocation.

Graph neural networks such as GCN and GAT show how node representations can be updated by relational neighborhoods [Kipf and Welling, 2017, Veličković et al., 2018]. These models are powerful when graph structure is informative, but standard benchmarks often assume fixed graphs and node labels. Unlike that setting, this project predicts continuous edge labels on daily crypto graph

snapshots. The hard and softmax GAT variants adapt attention to financial dependence: message passing is either restricted to strong historical correlations or biased toward high- $|\rho|$ neighbors.

Temporal graph models extend GNNs to evolving graphs [Xu et al., 2020, Rossi et al., 2020]. More recent transformer-based temporal GNNs such as DyGFormer emphasize learning node correlations and long temporal dependencies in dynamic graphs [Yu et al., 2023]. Finance graph work has used learned relations for stock return prediction [Li et al., 2020]. Those approaches focus mainly on representation learning or directional price movement. Here, the graph is a PIT liquidity universe, the label is future realized correlation, and evaluation includes both statistical error and economic utility through ERC and TSMOM allocation.

4 Methodology

Forecasting pipeline and labels. For every active PIT pair, I construct a daily edge row using only information available through date t . The base forecast is the trailing 60-day rolling correlation of overlapping 3-day returns, and the supervised label is the future realized correlation over $H \in \{60, 120\}$ days. I use 3-day log returns rather than raw daily returns to smooth day-to-day crypto noise, following the risk-forecasting idea that realized return aggregation can improve volatility/covariance measurement [Andersen et al., 2003]. To avoid immediate return overlap, the first future return used in the label starts after the current 3-day return window. All learned models predict the residual $y_{ij,t} = \rho_{ij,t}^{\text{fwd},H} - \rho_{ij,t}^{\text{base}}$ and then clip $\hat{\rho}_{ij,t}$ back into $[-0.99, 0.99]$.

Features. HGB, residual MLP, Edge GRU, and the all-node Transformer all use the same pair-level tabular features: three lags of 3-day returns for each coin, return spread and absolute spread, 7-day and 60-day volatility for each coin, volatility spread/ratio, baseline correlation, one-day baseline-correlation change, 5-day rolling mean of baseline correlation, absolute baseline correlation, and 60-day EWMA correlation. GAT and the all-node Transformer also separate information into node and edge channels. Node histories contain lagged 3-day returns, 7-day and 60-day volatility, and short/60-day momentum. Edge features contain baseline and EWMA correlation, correlation changes, return spread, and volatility spread/ratio.

Technical justification. The model ladder is designed to answer whether learning improves on common finance correlation baselines. Persistence and EWMA are standard, transparent risk-model baselines. Residual HGB is the strongest non-deep tabular model I considered because boosted trees handle nonlinear interactions among returns, volatilities, and correlation features with little preprocessing. Temporal GAT is motivated by the fact that crypto correlations are graph-structured: a coin’s covariance behavior may depend on which other coins it is currently linked to and how strongly those neighbors move together. Because day-to-day crypto correlations are noisy, the hard edge threshold and tuned softmax temperature τ are intended to denoise message passing by either filtering weak relationships or controlling how sharply attention concentrates on high- $|\rho|$ neighbors. I test both 60d and 120d realized-correlation targets, plus slow and fast TSMOM signals, to check whether results are robust across forecast horizons and signal speeds. Finally, TSMOM follows the time-series momentum literature [Moskowitz et al., 2012]; it is also a natural crypto test because momentum is a common trading signal and prior cryptocurrency evidence finds strong time-series momentum in returns [Liu and Tsyvinski, 2021]. The practical goal is not just lower correlation RMSE, but whether better covariance estimates let a trading signal realize higher out-of-sample Sharpe through better risk allocation.

Baselines and tree model. Persistence predicts ρ^{base} unchanged. EWMA replaces this with the 60-day EWMA correlation. Residual HGB fits a histogram gradient boosting regressor on pair features. The fixed HGB baseline trains on all 720 pre-test days with manually chosen stable settings. The tuned HGB uses the 600/120 train-validation split and selects max leaves in $\{15, 31, 63\}$ and minimum leaf samples in $\{30, 50\}$, with fixed learning rate 0.05, $L_2 = 0.05$, and 180 boosting iterations.

Deep learning baselines. I include two neural baselines to separate graph value from generic nonlinearity. The residual MLP applies two hidden layers to the same pair features as HGB, with no graph or sequence structure. The all-node Transformer encodes short node histories, applies full self-attention across active coins, and decodes pair residuals from node embeddings plus edge features. It tests broad node-level attention against GAT’s correlation-filtered message passing.

Temporal GAT. At each date, the active coins form a graph snapshot. A one-layer GRU encodes each coin’s 5- or 10-day node-feature history into a 32-dimensional state. Two GAT message-passing layers then update node embeddings. The hard GAT restricts messages to edges satisfying $|\rho^{\text{base}}_{ij}| \geq \kappa$, tuning $\kappa \in \{0.3, 0.5, 0.7\}$. The softmax GAT uses all active pair edges and adds a correlation-strength bias to the attention logits, $e_{ij} = a(Wz_i, Wz_j) + \tau|\rho^{\text{base}}_{ij}|$ and $\alpha_{ij} = \text{softmax}_j(e_{ij})$, with $\tau \in \{0.5, 2, 5\}$. The decoder MLP predicts residual correlation from $z_i, z_j, |z_i - z_j|, z_i \odot z_j$ and edge features. Dropout $\{0.10, 0.20\}$ and history length are chosen by validation.

Edge GRU. The Edge GRU is pair-centric rather than node-centric. For each pair, it feeds a sequence of pair features of length $\{30, 60, 90\}$ through a one-layer GRU with hidden dimension $\{32, 48\}$. It concatenates the resulting temporal embedding with learned asset/pair identity embeddings and predicts a residual through an MLP. This architecture tests whether explicit edge time series are more useful than graph message passing.

Training, ERC, and allocation. Neural models use Huber loss, AdamW, shuffled date snapshots during training, and validation-loss early stopping with a maximum of 20 epochs in the final notebook; validation selects both the configuration and the checkpoint epoch inside each fold. The ERC diagnostic converts each predicted correlation matrix into a covariance matrix $\Sigma = D_\sigma \rho D_\sigma$, solves long-only equal-risk-contribution weights, and computes utility loss relative to weights derived from realized future correlations. Downstream allocation uses slow TSMOM with lookbacks 21, 63, 262 and weights 0.125, 0.375, 0.5, and fast TSMOM with lookbacks 7, 21, 63 and weights 0.25, 0.35, 0.40. Raw views are clipped to $[-1, 1]$; predicted correlations and trailing volatilities form the covariance used to scale the views before applying them to next 3-day returns. No realized-vol overlay is used in the final report.

5 Experiments and Results

60d realized correlation						120d realized correlation					
Model	RMSE	Sp.	ERC	Slow	Fast	Model	RMSE	Sp.	ERC	Slow	Fast
Pers.	.242	.429	5.9	-.47	.72	Pers.	.221	.447	6.0	-.54	.72
EWMA	.231	.458	5.3	.18	.70	EWMA	.210	.469	5.8	.08	.70
HGB	.198	.437	2.3	.04	1.00	HGB	.162	.475	1.2	-.02	.83
HGB tuned	.201	.433	3.8	.07	.90	HGB tuned	.166	.464	2.0	.18	1.19
MLP	.204	.433	3.2	-.05	.88	MLP	.169	.468	2.1	.62	1.07
Transformer	.217	.424	3.1	.28	.81	Transformer	.178	.470	2.6	.45	1.27
GAT hard	.214	.444	3.9	.47	1.27	GAT hard	.176	.468	3.4	.58	1.29
GAT soft	.209	.444	3.6	.56	.86	GAT soft	.178	.467	2.3	.42	1.06
Edge GRU	.209	.406	3.6	.07	.74	Edge GRU	.176	.405	2.0	.05	1.10

Table 1: Main 60d/120d results. Sp. is mean daily Spearman; ERC is utility loss in bps; Slow/Fast are TSMOM Sharpes.

Experimental Setup. *Baselines:* persistence, 60-day EWMA correlation, fixed residual HGB, residual MLP, and all-node Transformer. *Proposed/ablated models:* tuned HGB, hard-threshold Temporal GAT, softmax- $|\rho|$ Temporal GAT, and Edge GRU. Hyperparameters are selected by chronological validation inside each fold: HGB tunes max leaves and minimum leaf samples; GAT tunes history length, dropout, and either threshold κ or softmax temperature τ ; Edge GRU tunes sequence length and hidden dimension; the MLP tunes hidden width; the Transformer tunes lookback length and dropout. Models are implemented in scikit-learn/PyTorch and run on local CPU/Mac with some neural trials on a T4 GPU. Full tuning is multi-hour/overnight; cached final analysis runs in seconds.

Quantitative Results. RMSE is appropriate because the label is continuous correlation. Daily Spearman tests whether each model ranks pairwise correlations correctly for allocation. ERC utility loss measures covariance usefulness, and slow/fast TSMOM Sharpe measures economic value. Table 1 compares all completed methods statistically and economically; Figure 3 focuses on downstream Sharpe for both 60d and 120d correlation targets. A key result is that learned models beat the finance baselines: HGB, MLP, Transformer, GAT, and Edge GRU all reduce RMSE relative to persistence and EWMA. Downstream, graph-structured and attention-based models perform best in Sharpe: the all-node Transformer and Temporal GAT strongly beat the baselines. The MLP performs well on RMSE and has one strong downstream result, but it is less robust; overall, models with graph structure perform best economically.

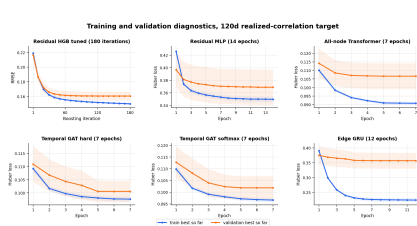


Figure 2: Selected 120d training diagnostics. HGB uses boosting-iteration RMSE; neural panels use epoch Huber loss.

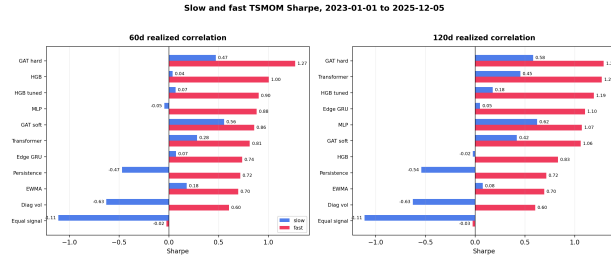


Figure 3: Slow and fast TSMOM Sharpe for 60d/120d forecasts. Hard-threshold GAT performs well in both fast experiments.

Which models capture the common mode when it matters most?

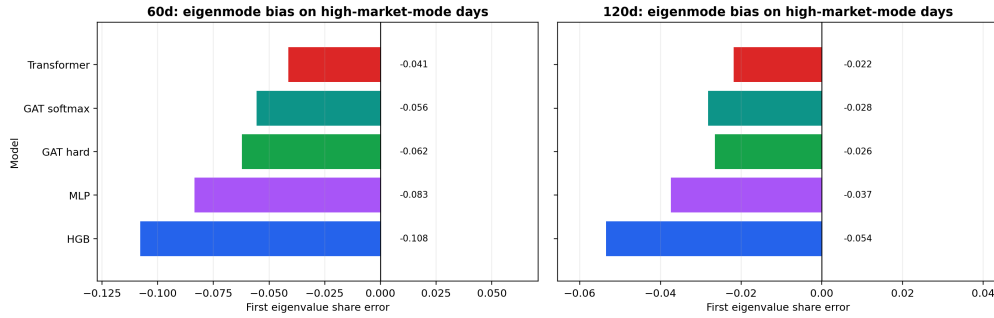


Figure 4: First-eigenvalue-share bias on high realized market-mode days. High market-mode days are forecast dates whose realized forward correlation matrix has a top-quartile first eigenvalue share: $\lambda_1/N \geq .7241$ for 60d labels and $\lambda_1/N \geq .7066$ for 120d labels.

Analysis and Discussion. *Qualitative analysis:* Figure 3 is an economic case study of the forecasts. In robustness terms, Temporal GAT is the strongest overall model: across both 60d and 120d realized-correlation labels, and across both slow and fast TSMOM signals, it improves substantially on persistence/EWMA pointwise baselines while remaining one of the best downstream allocators. Attention models dominate the fast TSMOM setting: hard GAT reaches 1.27 Sharpe with 60d labels and 1.29 Sharpe with 120d labels, versus about 0.70–0.72 for persistence/EWMA. Slow TSMOM is more mixed: GAT soft leads the 60d slow experiment, while MLP leads the 120d slow experiment. This suggests that the graph/attention models are useful on high-momentum dates even when they do not minimize average pairwise RMSE.

Deep-learning baselines: the residual MLP is a strong pointwise baseline: it is third-best by RMSE in both horizons (0.204 for 60d and 0.169 for 120d), showing that generic nonlinear learning helps. But its downstream fast Sharpe (0.88 and 1.07) still trails the best attention models, so lower pointwise error alone is not enough. The all-node Transformer is more interesting: its RMSE is weaker than HGB/MLP, yet it remains a strong fast allocator, especially at 120d where it reaches 1.27 Sharpe. This supports the idea that broad node-to-node attention can learn market-wide risk structure that matters for allocation, while GAT’s correlation-filtered attention is more consistently strong across horizons.

Temporal GAT sensitivity: the hyperparameter sweep supports the denoising motivation for graph attention. For the hard-threshold GAT, the strictest threshold $\kappa = .7$ has the best mean validation loss at both horizons and the best held-out test loss among thresholds: 0.160 for 60d versus 0.170/0.174 at $\kappa = .3/.5$, and 0.130 for 120d versus about 0.135 at lower thresholds. Longer 10-day node histories also improve held-out loss for both hard and soft GAT variants, suggesting that short temporal context helps separate persistent correlation structure from daily noise. The softmax- $|\rho|$ GAT is flatter: $\tau = 2$ is best for 60d, while $\tau = 5$ has the lowest 120d test loss. Overall, performance is strongest when the model either filters weak links or heavily weights high- $|\rho|$ neighbors, consistent with the idea that many low-correlation crypto edges add noise while high-correlation neighbors carry market-mode information.

Market-mode diagnostics: high market-mode days are the days where this first-eigenvalue-share diagnostic matters most: the crypto common market move is dominant, so underestimating the first eigenmode means underestimating the shared beta that drives portfolio risk. Figure 4 shows that HGB materially underestimates this high-market-mode bias, and the residual MLP also underestimates it despite having good pointwise RMSE. In contrast, models with access to broader graph structure—the all-node Transformer through full node attention and GAT through correlation-neighbor message passing—are much closer to the realized common-mode strength. Economically, this matters because a model that accurately estimates the crypto common market move can scale same-direction momentum exposure more appropriately in risk-on/risk-off regimes; in fast TSMOM, hard GAT has smaller max drawdowns than HGB at both horizons (27.4% vs. 29.5% for 60d, and 28.1% vs. 34.7% for 120d) with comparable realized volatility near 19–20%.

This is not only an unconditional-average effect: conditional on $|\hat{\rho} - \bar{\rho}| \leq .02$ (small average-rho forecast error, so mean-correlation error is not driving the result), HGB still understates the 120d first eigenvalue share on 79% of days; on the highest-realized-market-mode quartile, its first-eigenvalue bias is -.054 versus -.027 for hard GAT and -.022 for the Transformer. In crypto this may be useful: correlations are top-heavy and often rise together in volatile regimes, so a model with a small positive pairwise-correlation bias can still better reflect shared crypto beta for a same-direction momentum portfolio. This may explain why models with higher pointwise RMSE can still outperform downstream.

Error analysis: the model-specific failure modes are different. HGB is the strongest pointwise model, but its weakness is being too local: it fits pairwise residuals well while underestimating the market-mode eigenstructure that matters for portfolio risk. The residual MLP has a similar limitation; it learns nonlinear pair corrections, but without neighbor message passing it still misses part of the common crypto movement. Edge GRU underperforms because the PIT universe creates many sparse asset pairs, so pair embeddings have limited repeated history when hundreds of symbols rotate through only 50 active slots. Hard GAT is more robust downstream, but it can discard informative weak relationships when the threshold is too strict. Softmax GAT and the all-node Transformer can capture broader structure, but they may overstate broad positive correlation and sacrifice pointwise RMSE. Finally, all downstream results are before transaction costs, so high turnover in some TSMOM portfolios remains a practical weakness not captured by Sharpe alone.

Research questions: the proposal’s hypothesis was partly confirmed. Graph learning did not beat HGB on RMSE, but Temporal GAT is competitive pointwise, good on Spearman/ERC, and best or near-best downstream; all-node attention is an unexpectedly strong downstream baseline. More importantly, the graph neural models appear most valuable downstream, where they capture market-level correlation structure that models without broader structural understanding miss. The main conclusion is that correlation models should be judged both statistically and economically.

6 Conclusion

Taken together, the experiments suggest that crypto correlation forecasting is a portfolio-risk problem, not just an edge-regression benchmark. A model can fit average pairwise correlations very well while still missing the matrix structure that determines how much common risk a portfolio is carrying. In this setting, graph and attention models were most useful when correlations became market-wide: they acted less like universal RMSE improvements and more like detectors of shared crypto beta.

The practical lesson is to evaluate correlation forecasts at three levels: pointwise error, matrix structure, and downstream allocation. RMSE and Spearman remain important, but ERC utility and TSMOM Sharpe reveal whether the covariance forecast actually changes risk allocation in a useful way. The main limitations are still meaningful: the PIT universe creates sparse histories for rotating coins, turnover costs are excluded, and validation windows are short relative to crypto regime shifts. A stronger next version would train directly on covariance or utility-aware losses and evaluate transaction-cost-aware momentum allocation.

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